

**Human-Automation Interaction in Continuous Glucose Monitoring: A Formative Usability
Evaluation of AI-Generated Predictive Alerts in Adults 65 and Older**

Asha Golveo

Embry-Riddle Aeronautical University

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Dr. Tiziano Bernard

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Abstract

The integration of artificial intelligence into consumer health technology has introduced predictive alert features that require users to interpret and act on algorithmically generated forecasts rather than direct physiological measurements. The Dexcom G7 continuous glucose monitor (CGM) demonstrates this transition through its Urgent Low Soon alert, which predicts when a user's blood glucose will fall below 55 mg/dL within 20 minutes. Despite the implications for patient safety when predictive alerts are misunderstood, the literature on how adults aged 65 and older interpret and respond to these features is limited. This study conducts a formative usability evaluation of the Dexcom G7 companion application's predictive alert interface. Two participants completed a moderated, think-aloud usability session involving ten progressive static scenarios depicting glucose readings, trend arrows, predictive alerts, and signal loss screens. Pre-session and post-session surveys assessed baseline alert comprehension and trust in automation using the validated Trust in Automation scale (Körber, 2019). Participants showed strong comprehension across the majority of scenarios, with both participants correctly interpreting trend arrows, urgency coded alerts, glucose graphs, and signal loss screens. However, a comprehension gap was identified in the interpretation of the Urgent Low Soon predictive alert banner. Participants identified red color coding as the most visible cue across scenarios of high urgency. Although both participants demonstrated high reliance on predictive alerts regardless of trust scores, their trust ratings diverged based on their different levels of CGM experience.

Human-Automation Interaction in Continuous Glucose Monitoring: A Formative Usability Evaluation of AI-Generated Predictive Alerts in Adults 65 and Older

The rise of artificial intelligence (AI) in consumer health technology has introduced a new class of medical device features, requiring users to interpret and act on algorithmically generated predictions rather than direct measurements. Among the most consequential of these features is predictive analytics embedded in continuous glucose monitoring (CGM) systems with the ability to forecast glucose levels and identify trends. These systems are employed by manufacturers such as Dexcom, FreeStyle Libre, and Medtronic, with the goal of analyzing historical data to predict the onset of hypoglycemia. Unlike alerts that reflect a current physiological state, predictive alerts require users to understand and trust a forecast, thereby acting on that forecast before symptoms are present. With diabetes affecting approximately 40.1 million adults in the United States, including a disproportionately high prevalence among adults aged 65 and older, the risk of severe hypoglycemia can carry elevated risks of cardiac arrhythmia, cognitive impairment, and death (Centers for Disease Control and Prevention [CDC], 2026; Nakhleh & Shehadeh, 2021). For senior adults managing diabetes, the usability of predictive CGM alerts has direct implications for patient safety if users are unable to correctly perceive, comprehend, and respond to the alerts.

The Dexcom G7 is a CGM device that pairs with a companion smartphone application to provide real-time glucose readings, trend arrow indicators, and automated alerts. The G7 uses a machine learning algorithm to forecast when a user's blood glucose will fall below 55 milligrams per deciliter (mg/dL) within 20 minutes. This predictive alert is classified as software as a medical device (SaMD) under the U.S. Food and Drug Administration (FDA) regulatory framework. Its usability is subject to human factors engineering requirements outlined in FDA

Human Factors Engineering guidance (2016) and the international standard IEC 62366-1:2015, Usability Engineering for Medical Devices. Both frameworks require that use related risks, including use errors arising from failures of perception, comprehension, or decision making are identified and addressed through structured usability evaluation.

Despite the clinical importance of predictive CGM alerts, the human factors literature on how users interpret and respond to these features is limited. Furthermore, adults 65 and older are underrepresented in medical device usability research, despite being the primary users of many of these devices (Fox-Rawlings et al., 2018). This population presents significant human factors challenges relevant to the usability of predictive alerts, including variable smartphone proficiency, reduced vision and hearing, cognitive challenges that exacerbate cognitive load, and elevated rates of hypoglycemia unawareness in which physiological warning signs are absent or diminished (Isaković et al., 2016; Freeman, 2019; Wildenbos et al., 2018). Furthermore, research on human-automation interaction has established that both over-trust and under-trust in automated systems can produce inefficient use behavior, with over-trust manifesting as automation complacency and failure to critically evaluate alerts, while under-trust may manifest as alert dismissal or delayed response (Lee & See, 2004; Parasuraman & Manzey, 2010).

This study addresses these gaps through a formative usability evaluation of the Dexcom G7 companion application's predictive alert interface, focusing on adults aged 65 and older who are new device users. Participants were presented with ten static screenshots of the Dexcom G7 application, asked to interpret each display, and describe their intended response using a think-aloud protocol. Pre-session and post-session surveys assessed baseline and post-exposure comprehension of the "Urgent Low Soon" alert, in addition to measuring trust in the app's automated features using the validated Trust in Automation scale (TiA; Körber, 2019). A

structured moderator observation log captured use errors and close calls per critical task. The study was designed and conducted in accordance with FDA Human Factors Engineering guidance (2016) and IEC 62366-1:2015, with the goal of producing a formative evaluation and subsequent interface redesign recommendations.

Delimitations

This study is delimited to adults aged 65 and older who are currently using the Dexcom G7 and began using the device within 60 days prior to study participation. The eligibility criteria was established to ensure that participants represent new users whose initial comprehension and trust have not been substantially influenced by extended device experience. Furthermore, the study evaluates the Dexcom G7 companion app interface, and does not examine the sensor hardware or wearable device. These findings are also focused on the app's display of glucose readings, trend arrows, and alerts as presented on a smartphone screen. The usability evaluation is conducted using static screenshots rather than a live or simulated app session. This delimitation was imposed to standardize scenarios across participants, ensuring that all participants are exposed to the same interface regardless of their current glucose level or alert history. In addition, the study is conducted via remote Zoom video sessions rather than in person. This delimitation reflects the timeline constraints of this research, and the barriers of recruiting new, older adult users of CGMs.

Limitations

The most significant methodological limitation of this study is the use of static screenshots rather than a live application session. Participants are not interacting with a functioning CGM app on their own device, but viewing images of app screens on a shared Zoom display. This format will not capture the perceptual salience of real-time alerts, including audio

tones, haptic vibration, and push notification behavior which may influence perception and response. Behavioral intentions reported by participants in response to static images may not accurately reflect their actions in real-world alert situations. IEC 62366-1 acknowledges this limitation as inherent to the formative evaluation method, with simulated evaluations involving necessary departures from actual use conditions. Furthermore, participants will view screenshots on their own devices that vary in screen size, resolution, and brightness during the remote session over Zoom. However, these differences are representative of the variability of users who use different smartphones, versions of operating systems, display brightness, and text size.

Literature Review

Diabetes Management and Continuous Glucose Monitoring

The burden of diabetes disproportionately affects older adults, with 28.8% of adults aged 65 and older living with diabetes, nearly triple the rate observed in adults under 40 (CDC, 2026). Hypoglycemia, or dangerously low blood glucose, is a major barrier to achieving glycemic targets, with episodes that can range in severity from mild discomfort to loss of consciousness, seizure, and death (Nakhleh and Shehadeh, 2021). The risk of hypoglycemia is compounded in older adults by a phenomenon known as hypoglycemia unawareness, in which physiological symptoms such as tremor, sweating, and anxiety are muted or absent (Freeman, 2019). The risk of hypoglycemia unawareness among this vulnerable population necessitates that glucose monitoring technologies are timely, understandable, and accurate.

Continuous glucose monitors (CGMs) have emerged over the past two decades as an effective approach to diabetes management. Unlike traditional fingerstick blood glucose meters that provide only a single point in time measurement, CGMs use a small subcutaneous sensor to continuously measure interstitial glucose levels, transmitting data every few minutes to a

companion smartphone application or dedicated receiver. This continuous stream of data provided by CGMs enables users to observe their current glucose level, the direction and rate of change, and information that is essential for anticipating and preventing hypoglycemic and hyperglycemic events before they occur. The Dexcom G7 is a seventh-generation CGM system cleared by the U.S. Food and Drug Administration (FDA).

Garg et al. (2022) evaluated the accuracy and safety of the G7 in a 316-participant multi-site trial, reporting a mean absolute relative difference (MARD) of 8.2% for arm-placed sensors, demonstrating single-digit accuracy across a range of glucose levels and rates of change. Welsh et al. (2024) further compared G5, G6, and G7 systems, conducting a task analysis that found the G7 was favored over prior systems related to ease of use and positive clinical outcomes. The G7's companion app introduces AI-generated predictive alerts that can project where a user's glucose will be within a defined time window, thereby providing valuable information for diabetes management while also resulting in higher cognitive demands placed on users.

Senior Adults as a Priority User Population for CGM Technology

Adults 65 and older are a primary intended user group for CGM technology due to the high prevalence of diabetes among this population. However, the variability of older adults ranges from physical, cognitive, and experiential characteristics that may significantly influence their ability to effectively interact with digital health devices. These factors make the accurate comprehension of predictive low glucose alerts a concern for usability and patient safety. The MOLD-US framework, a model that systematically catalogues the barriers older adults face when using mobile health applications, identified age related changes in cognition, vision, motor function, and technology experience as key factors of interface design that can determine

usability outcomes (Wildenbos et al. 2018). Cognitive aging factors such as reduced processing speed, decreased working memory capacity, and greater difficulty interpreting novel or ambiguous information may impair an older adult's ability to rapidly decode a predictive alert and initiate an appropriate response. Isaković et al. (2016) found that mHealth apps specifically designed for or used by elderly populations have several recurring problems, including overly complex information displays, inconsistent terminology, inadequate feedback for user actions, and interface elements that older users may not be familiar with. The authors argue that the design of diabetes apps have rarely incorporated usability evaluations with older users, resulting in interfaces that are technically functional but practically inaccessible for this population. Furthermore, an analysis of diversity in medical device clinical trials found that older adults, women, and racial and ethnic minorities are systematically underrepresented in device validation studies (Fox-Rawlings et al., 2018). Recent reviews of CGM use in older adults confirm that devices offer clear clinical benefits, but usability challenges specific to this population remain underexplored.

Human Factors Engineering and Usability in Medical Device Design

Human factors engineering (HFE) is the scientific discipline concerned with understanding human capabilities and limitations and applying that knowledge to the design of systems, tools, and environments that support safe and effective human performance. In the context of medical devices, HFE has become a central regulatory requirement rather than an optional design enhancement. The FDA's foundational guidance document, "Applying Human Factors and Usability Engineering to Medical Devices" (2016), defines HFE as the application of knowledge about human behavior, abilities, limitations, and other characteristics to the design of medical devices. The FDA requires that manufacturers demonstrate the performance of

systematic evaluation, showing that users can operate their devices safely and effectively. The FDA's 2016 guidance distinguishes between formative and summative usability evaluations, a distinction that is central to the methodology of the present study. Formative evaluations are conducted iteratively during the design and development process to identify usability problems while design changes are still feasible. Summative evaluations are conducted on the finalized device to validate that use-related risks have been adequately mitigated.

Formative and summative evaluations are grounded in the concept of use-related risk, the recognition that many adverse medical device events are caused by use errors, and not by device malfunction. Use errors are actions or inactions by a user that produce a result different from what the user intended, or what the device manufacturer expected (FDA, 2016). FDA guidance emphasizes the importance of evaluating devices with representative users in realistic use environments, while noting that device designers cannot reliably anticipate the full range of user behaviors and misunderstandings without direct observation. The international standard IEC 62366-1 provides a complementary framework for usability engineering in medical devices, establishing a systematic process for identifying use scenarios, specifying user interface requirements, evaluating designs, and documenting usability evidence. The FDA's guidance and IEC 62366-1 constitute the professional standard of practice for HFE in the medical device industry.

Usability of CGM Devices and Companion Applications

Despite the rapid clinical adoption of CGM technology, the usability of CGM devices and their companion applications has received limited attention in the HFE literature, with research focusing on alert fatigue as the primary usability challenge. However, a narrative review of CGM use in older adults found that interpreting and responding to trend arrows, indicating the

direction and speed of glucose change, presents a particular challenge for the senior population (Price et al., 2024). The large amount of data generated by CGM devices can create confusion when older adults are confronted with continuous streams of numerical and graphical information. These findings are directly relevant to the present study, which evaluates comprehension of trend arrows as a specific use scenario. Welsh et al. (2024) included a task analysis in their comparison of Dexcom generations, finding that the G7 represented an improvement in ease of use over earlier systems. However, their task analysis focused on device setup and sensor application rather than app-based alert comprehension, leaving the cognitive demands of interpreting AI-generated predictive alerts largely unaddressed.

AI-Generated Predictive Alerts and User Comprehension

The G7's Urgent Low Soon alert utilizes predictive analytics to project when a user's glucose level will reach 55 mg/dL within 20 minutes, prompting the user to act on that projection before the event occurs. This type of predictive, time-bounded alert requires users to understand their current glucose value, as well as the probabilistic forecast, the meaning of the time window, and the appropriate behavioral response to a predicted rather than confirmed physiological state. Trust in automation is a foundational construct for understanding how users respond to AI-generated alerts. Lee and See (2004) provide the seminal theoretical framework, defining trust in automation as the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability. Trust in automation can be distinguished between appropriate and inappropriate reliance, whereby over-reliance occurs when users trust automated systems beyond their actual capabilities, while under-reliance occurs when users dismiss reliable automated information in favor of manual assessment (Lee and See, 2004). Both forms of miscalibrated trust represent safety risks in the context of CGM predictive alerts, with

over-reliance leading users to act on alerts without considering their situational context, while under-reliance may lead them to ignore or delay valid warnings.

Parasuraman and Manzey (2010) argue that automation can induce a passive monitoring mode in which users reduce their active engagement with a system, and become less likely to detect and correct automation errors. CGM users who habitually trust the device may accept alert information without fully processing its meaning, exhibiting a form of automation complacency in which the user relies on the CGM alert and ignores their own physical symptoms.

Synthesis and Research Gap

The usability of CGM companion applications, specifically the comprehension of AI-generated predictive alerts, has received minimal systematic HFE evaluation, and the limited research that does exist suggests that older adults face meaningful barriers in interpreting the information these devices provide. The automation trust literature establishes that user responses to AI-generated alerts are mediated not only by alert content but by users' mental models of the system, their calibration of trust in its predictions, and the degree to which they engage in active versus passive processing of automated information. There is a gap between the understanding of human-automation interaction and the practical reality of how older CGM users respond to the alerts. The present study addresses this gap through a formative human factors usability evaluation of the Dexcom G7 companion app with adults aged 65 and older, using a concurrent think-aloud protocol to evaluate comprehension of predictive alerts, trend arrows, and glucose readings. Consistent with FDA guidance on formative evaluation, this study aims to identify use errors and comprehension difficulties that can inform iterative design improvements.

Methodology

Research Approach and Design

This study employed a mixed methods research design, integrating qualitative and quantitative data collection and analysis within a single formative usability evaluation. Mixed methods were selected because the research questions required both the interpretive depth of observed task performance data, and the comparability of standardized scores across participants. The qualitative think-aloud protocol captures the individual participant's interaction with the Dexcom G7 interface, allowing the moderator to collect the words used by participants to describe what they perceived, the reasoning behind their interpretation of alerts, and use errors and close calls observed in real time. The quantitative component captures pre-to-post comprehension change scores and Trust in Automation (TiA) subscale scores. This approach was chosen to allow each component to compensate for the limitations of the other. The quantitative measures provided standardized, comparable scores that the qualitative think-aloud data alone could not produce, while the qualitative data provided the interpretive depth and contextual perspective that numeric scores cannot capture.

Design and procedures

The study was conducted in three sequential phases: pre-session, usability session, and post-session. Participants provided informed consent electronically before the pre-session survey and verbally at the opening of the usability session.

Phase 1: Pre-Session Survey

Eligible participants were sent a secure link to a pre-session survey administered through Qualtrics. The survey opened with eligibility screening questions; participants who did not meet all inclusion criteria were directed to a disqualification message and did not proceed further.

Eligible participants completed demographic and background items, technology comfort ratings, and three alert comprehension items constituting the baseline knowledge measure. Completion of the pre-session survey was required before a session appointment was confirmed.

Phase 2: Usability Session

Usability sessions were conducted remotely via Zoom video conferencing. At the opening of each session, the moderator introduced the study, reaffirmed the key elements of consent including the right to withdraw at any time, and confirmed verbal consent to audio recording. Participants were presented with ten static screenshots of the Dexcom G7 application via Zoom screen share, one at a time, in a fixed sequence. The fixed sequence was used to ensure that all participants were exposed to scenarios in the same order, and that the progression from lower-stakes to higher-stakes displays was consistent across sessions.

For each screenshot, the moderator followed a structured probe sequence derived from the moderator script. Perception probes were always delivered before comprehension and decision probes, in order to distinguish failures at the perception stage of the task chain from failures at later stages. The moderator did not provide corrective feedback, confirm or deny participant interpretations, or suggest actions during scenario tasks. Following completion of all ten scenarios, the moderator asked three standardized wrap-up questions addressing overall comprehension difficulties, moments of uncertainty, and participant suggestions for interface improvement. Sessions were audio recorded with participant consent.

Phase 3: Post-Session Survey

Immediately following the session, participants received a secure link to a post-session survey administered through Qualtrics. The survey assessed post-exposure alert comprehension using items parallel to the pre-session baseline, administered the full 19-item Trust in

Automation scale, and collected scenario-specific trust ratings, behavioral intention responses, and perceived usability ratings.

Apparatus and materials

Hardware

The moderator conducted all sessions from a single computer equipped with a built-in webcam and microphone, with Zoom used as the primary session audio recording tool. Participants joined sessions on their own personal devices, which varied in type and screen size.

Software

Both the pre-session and post-session surveys were administered using Qualtrics Survey Platform. Remote usability sessions were conducted using Zoom Video Communications for video conferencing, screen sharing of stimulus materials, and audio recording. Quantitative data were analyzed using Python.

Stimulus Materials

Ten static screenshots of the Dexcom G7 companion application served as the primary stimulus materials for the usability session. Screenshots were captured from active Dexcom G7 accounts and personally identifiable account information was removed.

Sample

This study employed a purposive non-probability sampling strategy, requiring participants with a precise combination of characteristics that could not be achieved through random selection from a general population. The target population for this study was adults aged 65 and older who were current Dexcom G7 users within 60 days before the usability session. Findings from this study cannot be statistically generalized to the broader population of older adult Dexcom G7 users due to the formative purpose of the study.

Eligibility Criteria

Inclusion criteria were: (1) age 65 years or older; (2) current diagnosis of diabetes mellitus confirmed by a healthcare provider; (3) at least one year of diabetes management experience; (4) current Dexcom G7 user who began using the device within the 60 days prior to participation; (5) ownership of a smartphone capable of running the Dexcom G7 application; and (6) ability to read and understand English. The criterion limiting participation to users within 60 days of first Dexcom G7 use was established to ensure that the sample represented new users whose comprehension and trust had not yet been substantially shaped by extended device familiarity, which is the use condition of greatest formative interest. There were no exclusion criteria based on diabetes type, gender, education level, or prior CGM experience with other devices.

Sample Size

A target sample of 8 to 12 participants was established, consistent with formative usability evaluation norms. Nielsen and Landauer (1993) demonstrated that five participants are sufficient to identify approximately 85% of usability problems within a homogeneous user group, and FDA Human Factors Engineering guidance (2016) endorses this benchmark for formative evaluation.

Sources of the Data

Data for this study were collected from four primary sources: the pre-session survey, the usability session, the moderator observation log, and the post-session survey. All data were generated through direct interaction with study participants and were collected prospectively during the study period. No archival data, secondary datasets, or existing databases were used as data sources.

Validity

Validity is measured using the dimensions of construct validity of the instruments and internal validity of the study procedures. The TiA scale demonstrates established construct validity through factor analyses conducted across multiple validation samples. The alert comprehension items were designed with direct reference to the clinical and technical definition of the “Urgent Low Soon” alert, ensuring that correct responses reflect genuine comprehension of the construct rather than general knowledge of hypoglycemia.

Several procedures were implemented to support internal validity. First, the use of a structured moderator script with standardized probe sequences ensured that all participants were asked equivalent questions in equivalent order, reducing variability in data extraction that could produce artificial differences across participants. Second, perception probes were consistently administered before comprehension probes, preventing the moderator from inadvertently anchoring participants’ interpretations through the sequencing of questions. Third, the moderator did not provide corrective feedback or evaluate participant responses during scenario tasks, preventing demand characteristics from shaping task performance. Fourth, audio recordings were reviewed against observation log entries to verify the accuracy of real-time documentation.

Treatment of the Data

Data Identification and Storage

All participants were assigned a unique participant ID code at the time of pre-session survey completion. This code was used to link pre-session survey data, session audio recordings, moderator observation logs, and post-session survey data across all three study phases without the use of any personally identifiable information. Participant names, contact information, and

identifying details were stored in a separate master list accessible only to the principal investigator.

Results

Participant Profiles

Two participants were recruited for this study, referred to as Participant 1 (P1) and Participant 2 (P2). Both were males aged 65–69 years, active Dexcom G7 users, and English-speaking. Table 1 summarises their key demographic and clinical characteristics.

Table 1. Participant Background Characteristics

Characteristic	Participant 1	Participant 2
Age	65–69	65–69
Gender	Male	Male
Diabetes Type	Type 2	Type 1.5 / LADA
Years Managing Diabetes	1–5 years	More than 20 years
Time Using Dexcom G7	~2–3 months	~2–4 weeks
Prior CGM Experience	None (G7 is first CGM)	Previously used Dexcom G6
Insulin Management	Diet/lifestyle	T-Slim insulin pump (integrated with G7)
Readings Per Day	4–6 times	7 or more times
Fingerstick Use	Yes – for cross-calibration	Yes – for calibration and cross-checking
Smartphone	Android; daily health app user	Uses laptop for Zoom; daily health app user
How Learned G7 App	Manual and doctor's office	Trial and error; uses Google and ChatGPT

The two participants differed in their depth of experience. P1 has been using the G7 for approximately two to three months and had no prior CGM experience, learning the app primarily through the device manual and guidance from his physician. P2 had more than twenty years of diabetes management experience, transitioning to the G7 from the Dexcom G6 approximately two to four weeks prior to the session. P2 also used a T-Slim insulin pump that integrates directly with the G7, creating a more technically complex use case. P2 described his approach to learning the G7 as trial and error, supplemented by Google searches and ChatGPT.

Pre-Session Alert Knowledge

Prior to the session, participants completed a background survey that included questions assessing their pre-existing conceptual understanding of the Dexcom G7's alert system. Table 2 summarises their responses to the three key alert-comprehension items.

Table 2. Pre-Session Alert Knowledge by Participant

Question	Participant 1	Participant 2
What does 'Urgent Low Soon' mean?	"You are going low soon" ✓	"I need to get something to eat or drink" ✓
Difference between predictive and current low alert?	Correctly distinguished: predictive = warning before; current = already low ✓	Correctly distinguished: predictive = warning before; current = already low ✓
Action when receiving 'Urgent Low Soon'?	Eat/drink fast-acting carbohydrates ✓	Eat/drink fast-acting carbohydrates ✓

Both participants demonstrated correct conceptual understanding of the 'Urgent Low Soon' alert and its distinction from the current 'Urgent Low' alert prior to engaging with any

scenarios. Both also identified the correct recommended response of consuming fast-acting carbohydrates, indicating that foundational alert literacy was present in both cases before the session began.

Scenario Observations and Think-Aloud Findings

The moderated session presented ten scenarios depicting different Dexcom G7 app screens, progressing from routine glucose readings to urgent alerts and signal loss. Participants were asked to think aloud while reviewing each screen and were probed by the moderator. Table 3 provides a summary of comprehension across all scenarios.

Table 3. Scenario Comprehension Summary

Scenario	Display	Participant 1	Participant 2
1	Steady arrow, 115 mg/dL	Correct: "Holding steady, good number" ✓	Correct: "Normal, I'm okay" ✓
2	Trending up arrow, 230 mg/dL	Correct: identified rising trend, described rate of change ✓	Correct: checked insulin on board, monitored pump ✓
3	Trending down arrow, 154 mg/dL	Partial: recognised downward trend but recommended exercise rather than watchful waiting 🟡	Correct: "Not urgent at 154, I'd leave it alone" ✓
4	Straight down arrow, 197 mg/dL	Correct: compared arrow speed vs. Scenario 3, identified faster descent ✓	Correct: "Going down quicker, hold tight and watch" ✓

5	Double down arrows, 46 mg/dL, red	Correct: "Red really stands out, hurry up and do something" ✓	Correct: "Get up and drink apple juice, going down rapidly" ✓
6	Graph only, 3-hour trend	Correct: read threshold lines, described trend direction accurately ✓	Correct: identified upper/lower ranges, narrated glycaemic history ✓
7	Reading + graph, 156 mg/dL	Correct: connected reading to graph, noted signal gaps ✓	Correct: identified signal loss gaps, correctly linked reading to graph ✓
8	Urgent Low, 52 mg/dL, red, banner	Correct: immediate recognition of urgency, acted on red and banner ✓	Correct: "Red. 52 red. Get sugar in me." ✓
9	Urgent Low Soon, 58 mg/dL, banner + pop-up	Partial: understood urgency and acted appropriately; misread '55 mg/dL' as rate rather than predicted threshold ●	Correct: "Projecting it will be 55 within 20 minutes" ✓
10	Signal loss screen	Correct: "Lost signal, phone too far away" ✓	Correct: "Out of range, Bluetooth; manually check" ✓

Note. ✓ = *Correct*; ● = *Partial*; ✗ = *Incorrect*.

Trend Arrow Comprehension (Scenarios 1–4)

Both participants accurately interpreted all four trend arrow scenarios. In Scenario 1 (steady arrow, 115 mg/dL), both immediately identified the reading as stable and within a safe range requiring no action. In Scenario 2 (diagonal up-right arrow, 230 mg/dL), both participants recognised the upward trend and correctly identified it as elevated. However, their responses differed based on clinical context with P1 describing walking or drinking water to bring levels down, while P2 drew on his insulin pump to consider whether a bolus (a dose of insulin taken to handle a rise in blood glucose) was needed.

In Scenarios 3 and 4, which compared a gradual diagonal downward arrow (154 mg/dL) against a faster straight downward arrow (197 mg/dL), both participants correctly identified the directional difference in rate of change. P2 explicitly noted that the straight-down arrow was "going down quicker, at a more rapid pace," and both participants correctly withheld action given the values were still within manageable range. P1's response to Scenario 3 included recommending light exercise to manage the decline, slightly deviating from a waiting approach.

Urgent Alert Recognition (Scenarios 5, 8, and 9)

Scenario 5 (double down arrows, 46 mg/dL, red background) elicited immediate recognition from both participants. P1 focused on the red background as the primary alerting signal, stating that "the red really stands out" and associating it with danger. P2 similarly responded to the red color first, saying "I better get up out of bed and go drink some apple juice." Both correctly identified the double arrows as indicating a rapid rate of decline, and both described taking immediate corrective action.

Scenario 8 (Urgent Low banner, 52 mg/dL, red UI, 24-hour erratic graph) produced the same high-urgency response. Both participants noticed the red color first, correctly read the banner text aloud when prompted, and described actions to raise blood sugar immediately.

Scenario 9, the Urgent Low Soon predictive alert (58 mg/dL; banner: '55 mg/dL within 20 minutes'; pop-up: 'Urgent Low Soon, Act now to prevent low'), produced a divergence between the two participants. P2 correctly interpreted the alert, identifying 55 mg/dL as the projected glucose value and 20 minutes as the estimated time frame for that threshold to be reached. P2 stated "it's projecting a 20-minute time frame that's going to continue to drop to 55." P2 also stated that he would likely feel symptoms before the alert fired, and would manually

check if he did not. P1, by contrast, understood the urgency of the alert and planned to take corrective action, but misread the banner semantically. When P1 was asked what "55 mg/dL" referred to, he responded that "readings are going lower, dropping real fast, dropping more than 55 real quick", interpreting 55 as a rate of drop rather than a predicted glucose threshold. P1 described 20 minutes as "dropping real fast within 20 minutes" rather than as a predictive time window.

Graph Comprehension (Scenarios 6 and 7)

Both participants demonstrated solid comprehension of the glucose trend graphs. In Scenario 6 (graph only, three-hour range), both correctly identified the upper and lower threshold lines as the high and low alert boundaries, respectively. P1 described the red line at 75 as representing an urgent low, and the upper boundary at 200 as the dangerous high threshold. P2 similarly noted 200 as his upper range and 75 as his low range.

In Scenario 7 (156 mg/dL reading with diagonal down-left arrow and three-hour graph below), both participants correctly linked the current reading to the graph and identified the gaps in the dotted line as signal loss periods. P2 elaborated on this by specifying intermittent signal loss between approximately 6:45 and 8:30 based on the graph, and correctly inferred that the current reading of 156 mg/dL corresponded to the rightmost data point on the graph. Neither participant expressed confusion in this scenario.

Signal Loss Comprehension (Scenario 10)

Both participants correctly understood the signal loss screen. P1 identified it as the phone being too far from the sensor, noting that it sometimes takes up to thirty minutes to restore the signal. P2 associated the icon with a Bluetooth disconnection and indicated that he would

manually check his glucose while waiting for the signal to return. P2 also noted that the G7 tends to lose signal intermittently, and that it generally reconnects without intervention. Both participants indicated that the signal loss screen, rather than undermining trust, reinforced their understanding of how the system works.

Post-Session Survey Findings

Alert Comprehension Items

The post-session survey revisited alert comprehension using open-ended and multiple-choice questions. Both participants correctly identified the distinction between 'Urgent Low Soon' and 'Urgent Low' in their post-session responses, consistent with their pre-session survey answers. P1 described 'Urgent Low Soon' as meaning "going low really fast," which is behaviourally appropriate, but continues to reflect a threshold-based interpretation. P2's post-session response to the '55 mg/dL within 20 minutes' item was fully correct, stating that "It is an estimate of time of how long it will be before my glucose level falls to 55." P1's response to the same item was "going down fast", reflecting an imprecise understanding of the predictive threshold.

Trust and Reliability

The post-session survey included a multi-item trust scale covering app reliability, predictability, familiarity, and developer benevolence. Table 5 presents a comparative summary of key items.

Table 5. Post-Session Trust and Reliability Survey Responses

Trust/Reliability Dimension	P1	P2	Pattern
App interprets situations correctly (Q5)	Agree	Somewhat agree	↓ P2 more cautious
App performs tasks reliably (Q6)	Agree	Neither agree nor disagree	↓↓ Gap
App always provides accurate glucose info (Q8)	Agree	Neither agree nor disagree	↓↓ Gap
Easy to understand how app works (Q11)	Strongly agree	Neither agree nor disagree	↓↓↓ Large gap
Understand app well enough to predict behaviour (Q13)	Strongly agree	Somewhat agree	↓ Minor gap
Very familiar with app and its features (Q16)	Strongly agree	Neither agree nor disagree	↓↓↓ Large gap
Developers concerned about user wellbeing (Q17)	Strongly agree	Neither agree nor disagree	↓↓↓ Large gap
Confidence in alert accuracy — with symptoms (Q24)	Fairly confident	Fairly confident	= Same
Would act on alert without symptoms (Q25)	Completely confident	Fairly confident	↓ Minor gap
Rely on predictive alerts without HCP (Q31)	Completely	Completely	= Same

P1 reported notably higher trust across nearly all dimensions than P2. P1 responded with 'Agree' or 'Strongly agree' to all reliability and familiarity items, and 'Strongly agree' to all

developer benevolence items. P2's responses were more moderate, frequently selecting 'Neither agree nor disagree' or 'Somewhat agree,' particularly on items related to app predictability and familiarity. This difference is likely attributable to their different onboarding contexts. P1 had approximately two to three months of experience with the G7 following a structured introduction via manual and physician guidance, while P2 had only two to four weeks of experience and was still calibrating his understanding of the G7's differences from the G6.

Despite this difference in perceived reliability and familiarity, both participants indicated they would rely on the G7's predictive alerts 'completely' to manage a low blood sugar episode without involving a healthcare provider. Both also indicated they would act on the alert even in the absence of physical symptoms, and both selected 'Fairly confident' in the accuracy of the 'Urgent Low Soon' prediction. This suggests that both participants demonstrated a high level of reliance on the predictive alert.

Behavioural Reliance and Fingerstick Use

Both participants reported a reduction in fingerstick blood glucose testing since beginning the Dexcom G7, selecting 'I use fingerstick testing much less often than before.' However, both participants also described ongoing use of fingerstick testing for specific purposes such as cross-calibration when a reading seemed inconsistent, or verification when the CGM reading did not match their physical symptoms. This pattern of selective verification was consistent with their think-aloud responses during the session, where both participants mentioned checking a manual reading if the alert did not align with how they felt. When asked how likely they were to rely mainly on the G7 rather than a finger prick when responding to a low glucose alert, P1 selected 'Somewhat likely' and P2 selected 'Neither likely nor unlikely,' both reflecting a

measured rather than absolute reliance posture. Neither participant reported feeling they had over-relied on the app during the session.

Discussion

The most prominent finding of this evaluation is that both participants demonstrated strong overall comprehension of the Dexcom G7 interface across the majority of scenarios. Trend arrows, the urgency associated with the 'Urgent Low' alert, graph reading, and signal loss recognition were all correctly interpreted without observable difficulty. This is a meaningful finding given that both participants fell within the 65–69 age group, a demographic sometimes associated with lower technology proficiency, and that the app interface involves the interpretation of visual elements such as numeric readings, directional arrows, color coding, and banner overlays. However, a comprehension gap was identified in Scenario 9, the 'Urgent Low Soon' predictive alert, with P1 consistently misinterpreting the banner '55 mg/dL within 20 minutes' as indicating a rate of descent rather than a predicted future glucose threshold. This error was observed both during the think-aloud session and confirmed in the post-session survey, where his response to the same item remained 'going down fast', exhibiting a knowledge gap rather than a slip or simple misread. P1 understood the alert as urgent and responded appropriately in terms of action, but his mental model of what '55 mg/dL' represented was inaccurate. This distinction matters because a user who believes '55 mg/dL' refers to a rate of decline may not understand that their blood glucose is predicted to reach a dangerously low absolute threshold.

In situations where a user's current reading feels relatively safe, misunderstanding the predictive nature of the alert could lead to underestimation of urgency or delayed action. P2 did not exhibit this comprehension error. His correct interpretation “it's projecting a 20-minute time

frame that's going to continue to drop to 55", likely reflects his prior experience with the Dexcom G6 which also uses predictive alerting. P2 also has a more intensive use pattern of seven or more readings per day. This contrast between participants suggests that the comprehension gap may be more prevalent among newer or less experienced CGM users, though the sample is too small to generalise.

Color Coding Was the Most Salient Visual Cue

Across Scenarios 5, 8, and 9, both participants consistently reported the red background as the first and most immediately noticed element. In Scenario 5, P1 stated "the red really stands out" and P2 responded with "I better get up out of bed." In Scenario 8, both verbatim responses identified red before the numeric reading or banner text. This pattern aligns with established human factors principles regarding the use of color as a pre-attentive visual feature. Red signals danger reliably across contexts, including for older adults who may be slower in visual processing of complex displays. However, it is notable that the 'Urgent Low Soon' alert in Scenario 9 does not use the same full-screen red background as the 'Urgent Low' alert, despite representing a clinically serious predictive event. The less alarming visual presentation of Scenario 9 may have contributed to P1's lower urgency rating, and potentially reinforced his partial miscomprehension by not fully communicating the gravity of the impending situation. This raises a human factors design question about whether the visual hierarchy between the predictive and concurrent low alerts is appropriately calibrated.

Trust Was High but Differentiated by Experience

The post-session trust data revealed a meaningful divergence between the two participants that cannot be attributed to comprehension performance alone. P1's higher trust

ratings across nearly all dimensions of reliability, predictability, familiarity, and developer benevolence reflect both his more positive onboarding experience, and his longer period of G7 use. P2's more cautious, moderate trust is consistent with his transitional state of only having two to four weeks with the G7 after years on the G6. He was actively recalibrating his expectations and encountered calibration difficulties in the days immediately preceding the session. Despite this difference in generalised trust, both participants expressed high reliance on the predictive alert, with both indicating they would act on the 'Urgent Low Soon' alert even without physical symptoms, and both would rely on it completely to manage a hypoglycaemic episode without involving a healthcare provider. This high reliance, combined with P1's documented miscomprehension of the alert mechanism, represents a potential safety concern. A user who acts on an alert they do not fully understand may act appropriately (eating fast-acting carbohydrates) in most cases, but their decision-making is not grounded in an accurate understanding of what the alert is predicting. P2's additional safety check of manually verifying the reading if he did not feel symptoms reflects a more active calibration strategy that may mitigate over-reliance. P1 also mentioned fingerstick cross-checking in certain situations, though his language around this was less systematic.

Usability Heuristic Analysis

The findings can be contextualised within Nielsen's ten usability heuristics. Two heuristics are particularly relevant to the comprehension gap identified:

Match between system and the real world

The '55 mg/dL within 20 minutes' banner uses clinical notation that may not intuitively convey a predictive glucose threshold to users without a strong number or prior CGM

background. The format presents '55 mg/dL' as the primary figure, which users may interpret as a measure of change rather than a future value due to other readings throughout the app (e.g., the current reading of 58 mg/dL) displaying the same format. A more explicit framing, such as 'Your glucose may reach 55 mg/dL in 20 minutes,' could better align the interface with the user's existing mental models.

Help and documentation

Neither participant reported accessing educational content within the app about the predictive alert system. P1 learned the app primarily via the device manual and physician instruction, with his knowledge of predictive alerting correct at the conceptual level, but not at the level of banner interpretation. This suggests that available educational resources, while sufficient for basic alert response, may not provide adequate support for understanding the specific interpretation of the banner text.

Limitations

This evaluation was subject to several limitations that must be acknowledged, with the most significant being that only two participants were recruited, negating any generalisation of the findings. While this sample size is appropriate for the formative design of the study, the validity of the findings would be strengthened by a larger and more diverse sample. Participants with lower technology literacy, different diabetes types, or no prior CGM experience would provide important contrasting data. In addition, the evaluation was conducted remotely via Zoom. While this enabled participation from geographically distributed users, it may have introduced observer effects and reduced the moderator's ability to observe non-verbal cues. Furthermore, the scenarios were static images rather than a live or interactive application.

Participants could not tap, navigate, or dismiss alerts, which may have affected their engagement with certain elements. In particular, the pop-up modal in Scenario 9 may have been perceived differently in a real interaction context where the user would need to actively dismiss it. Lastly, both participants were male and within the same age bracket, limiting the demographic diversity of the sample. Future evaluations should recruit participants across a broader age range and include female participants, as well as users with varying levels of health literacy.

Conclusion and Recommendations

This formative usability evaluation examined the comprehension of the Dexcom G7 companion application's predictive alert system among two senior adult users. The study utilized a moderated think-aloud protocol with ten progressive scenarios, supplemented by pre-session and post-session surveys. The evaluation found that overall usability was high, with both participants successfully interpreting trend arrows, urgency-coded alerts, glucose graphs, and signal loss screens with minimal difficulty. The red background associated with the 'Urgent Low' alert was consistently identified as the most immediately salient design feature, and contributed to rapid urgency recognition. However, the evaluation identified a meaningful comprehension gap in the interpretation of the 'Urgent Low Soon' predictive alert's banner text. One participant consistently misread '55 mg/dL within 20 minutes' as indicating a rate of glucose decline rather than a predicted future glucose value, despite correctly understanding the alert's general meaning and taking appropriate action. This gap was not observed in the participant with more overall CGM experience, suggesting an experience or familiarity effect. This finding is significant, given that both participants expressed willingness to rely on predictive alerts without involvement from a healthcare provider. The evaluation also documented a notable divergence in generalised trust between participants, which may be attributable to differences in experience and onboarding

context. However, this did not translate into a difference of reliance on the predictive alert. The findings support the value of user-centred evaluation of medical device companion applications for older adult populations who may bring varied technology experience.

Design Recommendations

Recommendation 1: Revise the predictive alert banner to make the threshold prediction explicit.

Participant 1 misinterpreted '55 mg/dL within 20 minutes' as a rate of decline rather than a predicted threshold. The current format is ambiguous for users without strong CGM familiarity. The banner should be revised to more explicitly communicate the predictive nature of the value, such as 'Glucose predicted to reach 55 mg/dL in ~20 min'. This phrasing reflects natural prediction language, and makes the directionality and destination of the prediction transparent. Severity: High.

Recommendation 2: Increase the visual urgency of the 'Urgent Low Soon' alert to better communicate its predictive severity.

The 'Urgent Low Soon' screen does not use the same full screen red background as the 'Urgent Low' alert, which may inadvertently suggest lower severity. Consideration should be given to strengthening the visual urgency of the predictive alert such as a more prominent color treatment, animation, or a visual countdown element. Severity: Moderate to High.

Recommendation 3: Provide brief in-context tooltips or educational overlays explaining predictive alert semantics.

The first time a user receives an 'Urgent Low Soon' alert, an in-app tooltip or first-alert overlay explaining what the threshold value and time window mean could directly address the

gap in banner interpretation. This aligns with the 'help and documentation' heuristic and the principle of just-in-time learning for medical device users. Severity: Moderate.

Recommendation 4: Evaluate whether the alert terminology 'Urgent Low Soon' is the most comprehensible phrasing for non-expert users.

While both participants correctly identified the general meaning of the alert, the banner format did introduce semantic confusion. Alternative phrasings should be explored in future evaluations, such as 'Low Predicted in 20 Minutes', tested with a broader sample including first-time CGM users and lower health literacy populations. Terminology testing should be integrated into the next iteration of usability evaluation. Severity: Low to Moderate.

Recommendations for Future Evaluation

This formative evaluation should be considered a foundation for further iterative testing rather than a final assessment of the application's usability. A larger sample of at least five to eight participants is recommended for the next round of formative testing, with particular attention to recruiting users with less than one month of CGM experience, lower technology literacy, and diverse diabetes management approaches. Future evaluations should test the revised banner text (Recommendation 1) and increased visual urgency treatment (Recommendation 2) directly, using A/B scenario comparisons to assess whether the changes resolve the comprehension gap identified in this study. The inclusion of female participants and users across a broader age range (including those over 75) would strengthen the validity of findings for the target population of older adults with diabetes. Future studies should consider an in-person or think-aloud protocol using a working prototype or the live application, rather than static scenario images, to better capture interaction behaviours such as alert dismissal and navigation.

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